Predicting Coronary Heart Disease from Lifestyle Survey Data with Gradient Boosting

Marie Berbichashvili and Nafi Anna Dramé
EPFL, Switzerland

***Abstract*—Cardiovascular diseases are a leading cause of death, and identifying people at risk from survey answers alone could enable cheap, large-scale screening. We predict self-reported coronary heart disease or myocardial infarction (MICHD) from 321 survey answers of 328,135 respondents of the 2015 BRFSS survey, where only 8.83% are positive and many values are missing or coded. Tuning the decision threshold alone lifts the six linear baselines from an F1 score of about 0.1 to 0.434. A NumPy gradient boosting model with categorical splits and learned missing-value directions reaches 0.440 on validation and 0.448 on the test set. A cardiac-rehabilitation question, asked only after a heart attack, leaks the label and was the model's second most used feature; removing it lowers F1 by only 0.009 on validation and 0.006 on the test set, so the performance comes from genuine risk factors rather than leakage.**

I. Introduction

Cardiovascular diseases caused an estimated 19.8 million deaths in 2022 [1]. Common risk scores such as Framingham require a blood test, yet replacing cholesterol with body-mass index gives the same accuracy (c statistic 0.83 for women and 0.78 for men) [2]. This motivates predicting heart disease from survey answers alone. We use BRFSS 2015 [3], a U.S. telephone health survey with 328,135 training and 109,379 test respondents described by 321 features. A respondent is positive if a health professional told them they had coronary heart disease, a heart attack or angina.

Our contributions are: (i) showing that threshold tuning is essential for linear models under class imbalance; (ii) a NumPy gradient boosting model with categorical splits and learned missing-value directions, with an ablation of its components; and (iii) a leakage audit showing that a post-diagnosis question inflates feature importance but barely affects performance. Our code is available at https://github.com/marieberbipynb/EPFL-CS433-Project1.

II. Data and Exploratory Analysis

Class imbalance. Only 8.83% of respondents have MICHD, so always predicting "no" reaches 91.2% accuracy. We therefore evaluate with the F1 score of the positive class.

Feature groups. Features are survey administration variables (date, state, sequence number, sampling weights), raw answers, and variables computed by the CDC from these answers. We remove identifiers and survey weights, which describe the interview rather than the respondent. Computed variables often duplicate raw ones (height in inches and meters, $r = 0.99$), and duplicates share their importance in any model.

Missing values are informative. 30 features are missing for over 99% of respondents, because some modules are asked only in some states or only after a given answer. For instance, `BPMEDS` (blood-pressure medication) is asked only to respondents with high blood pressure: the 59.8% who did not answer have a MICHD rate of 3.8%, against 16.3% for those who did. Such skip patterns can also leak the label (Section IV).

Risk factors. The MICHD rate rises from 0.5% at 18–24 years to 22.2% above 80, and from 2.1% for "excellent" to 31.9% for "poor" general health. It is also higher for men, inactive respondents, and those with high blood pressure, high cholesterol or diabetes (Fig. 1). Former smokers have a higher rate than current smokers, likely because many quit after a cardiac event: these associations are not causal.

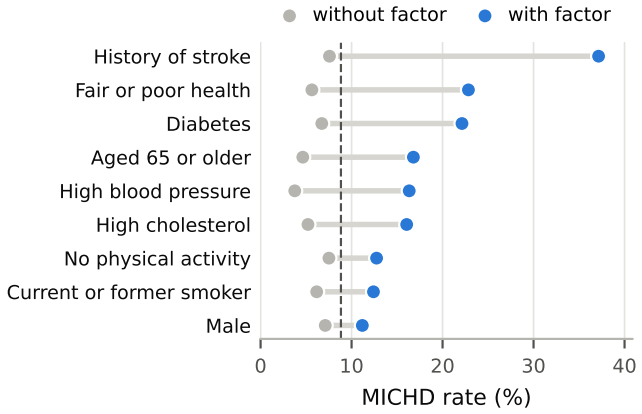


Fig. 1. MICHD rate with and without each risk factor (training set). The dashed line is the overall rate (8.8%).

III. Models and Methods

Protocol. All models are compared on a fixed stratified 80/20 train/validation split (seed 123), with preprocessing statistics computed on the training part only. The final model is retrained on the full training set before predicting the test set.

Linear baselines. For the six methods of `implementations.py`, we drop identifiers, survey weights and features missing for over 99.5% of respondents. Features with at most 20 distinct values are one-hot encoded with a missing category, which also absorbs special answer

codes (e.g. 7 "don't know", 9 "refused"). For continuous features, such codes are set to missing, then values are median-imputed, clipped to the 1st–99th percentiles and standardized, with a missing indicator when over 0.5% are missing. To capture age-dependent effects, we add one-hot interactions between age group and 11 risk factors, giving 2,252 features. Regression models use $y \in \{-1, 1\}$ and logistic models $y \in \{0, 1\}$, with $\lambda = 0.01$ (ridge) and 10^{-4} (regularized logistic); gradient methods run 500 iterations (50,000 for SGD).

Gradient boosting. Trees capture interactions and need neither one-hot encoding nor imputation, so the model uses the raw features after removing identifiers, survey weights and the leaky `HAREHAB1`. Each feature is mapped to at most 32 ordered bins (one per value, or quantiles), plus a bin for missing values. The model minimizes the logistic loss with Newton updates: respondent n has gradient $g_n = c_n(y_n - p_n)$ and hessian $h_n = c_n p_n(1 - p_n)$, with class weight $c_+ = 1.5$ for positives and 1 otherwise. Depth-3 trees are grown greedily with the split gain

$$\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda}, \tag{1}$$

where G, H sum g_n, h_n over a node and $\lambda = 20$; leaves predict $G/(H + \lambda)$. We add 250 trees with learning rate 0.06, at least 300 respondents per leaf and 160 random features per tree. For categorical features (at most 20 values), bins are sorted by G/H before searching the cut, so any grouping of categories can be found. For the others, the tree also tries sending missing values left, learning where they belong instead of imputing them. Two seeds are averaged after standardizing their scores, and the 12.7% highest-scoring respondents, a rate selected on validation, are predicted positive.

IV. Results

TABLE I
F1 OF THE LINEAR METHODS AND OF GRADIENT BOOSTING (GB). GB PREDICTS THE TOP 12.7% AS POSITIVE INSTEAD OF USING A THRESHOLD.

Method	Val. ($t = 0$)	Val. (tuned t)	Test
Least squares	0.143	0.434	–
MSE GD	0.007	0.424	–
MSE SGD	0.331	0.402	–
Ridge regression	0.091	0.434	–
Logistic regression	0.107	0.410	–
Reg. logistic regression	0.107	0.410	–
GB with `HAREHAB1` (leaky)	–	0.449	0.454
GB, final	–	0.440	**0.448**

With $t = 0$, linear models trained on imbalanced labels predict almost only negatives (Table I). Tuning the threshold alone brings ridge regression from 0.091 to 0.434; closed-form methods beat gradient-based ones, which have not converged after 500 iterations. Gradient boosting improves on the best linear baseline by only 0.6 points on validation (0.440 vs 0.434). On validation, the final model has a precision of 0.373, a recall of 0.536 and an accuracy of 0.879: it detects about half of the cases, at about 1.7 false alarms per true positive. Its test F1 (0.448) and accuracy (0.882) are close to these estimates.

TABLE II
VALIDATION F1 WHEN REMOVING ONE COMPONENT FROM A SINGLE BOOSTING MODEL. ABLATIONS WERE RUN WITH `HAREHAB1` INCLUDED.

Configuration	Validation F1
Full single model	0.449
without class weights ($c_+ = 1$)	0.447
without categorical splits	0.446
without learned missing direction	0.448
keeping identifier and weight columns	0.446
tuned threshold instead of top 12.7%	0.450

Each component brings at most 0.3 points (Table II); since these values come from a single validation split, such small differences may not be significant, and no single component clearly drives the performance. The 2-seed average does not change F1, and the top-12.7% rule matches a tuned threshold.

Label leakage. Counting the tree nodes that split on each feature, the model with `HAREHAB1` used it for 5.0% of splits, second only to general health (5.4%). Yet this question, on outpatient rehabilitation after a heart attack, is asked only to respondents who report one, which is part of the MICHD definition: it encodes the label rather than a risk factor, and would not exist for a person being screened. Removing it lowers F1 by only 0.009 on validation (0.449 to 0.440) and 0.006 on the test set (0.454 to 0.448). Without it, the most used features are age (`_AGEG5YR` and `_AGE80`, 8.0%) and general health (5.3%), followed by sex (4.3%), history of stroke (3.5%) and joint pain (2.8%), consistent with the risk factors of Section II; with the fitness estimates `MAXVO2_` and `FC60_` (4.1%), derived from age and sex only, age-related features reach 12.1% of splits.

V. Discussion

The leakage audit shows that feature importance can be misleading even when the score is not: the leaky question ranked second, yet the model barely needs it, because age, general health and comorbidities carry the same signal. The leaky model would also rank higher on the leaderboard; we keep the final one because only it is valid for screening. The gain over the best linear baseline is small and costly (20 to 40 minutes of training per boosting model, against about a minute for ridge regression), and ablations differ by at most 0.3 points, so cross-validation would be needed to compare them reliably. Split counts reflect associations, not causes: general health may be a consequence of heart disease. Finally, labels are self-reported and cross-sectional, so we detect existing disease rather than predict future events, and the top-12.7% rule assumes a similar test prevalence..

VI. Summary

Threshold tuning is the key step for linear models, and a NumPy gradient boosting model reaches a test F1 of 0.448 without the leaky feature, only 0.006 below the leaky model.

REFERENCES

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